

MATH 518 – Fall 2025

Prerequisites: A background in algebra at the level of MATH 456

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Office Hours: Tuesdays at 4:30pm-5:30pm in BURN 1120.

Course Topics: Radical ideals and Hilbert's Nullstellensatz. The Zariski topology. Irreducible decomposition. Dimension. Tangent spaces, smoothness and singularities. Projective spaces and projective varieties. Regular functions and morphisms. Rational maps and indeterminacy. Blowing up. Divisors and linear systems. Projective curves.

Course Materials: We will mainly follow Fulton's notes *Algebraic curves: An Introduction to Algebraic Geometry* that are freely available online. Other useful references for this course are

1. *An Invitation to Algebraic Geometry*, by Smith-Kahanpää-Kekäläinen-Traves
2. *Basic Algebraic Geometry 1*, by Shafarevich
3. *Algebraic Geometry: An Introduction*, by Perrin
4. Lecture notes of Eyal Goren and Brent Pym (see on MyCourses)

Means of Assessment: Your course grade will be the **maximum** of the following two assessment options:

Option 1: 30% assignments + 30% midterm + 40% final exam

Option 2: 30% assignments + 70% final exam

Assessment Type	Description	Due Date	Considerations and Late Penalties
Midterm Exam	In-person; Crowdmark Long answer, proof-based questions	In person during class time (4:05 – 5:25) on Monday, October 20.	Students missing the midterm for any reason will automatically be assessed by assessment option 2
Final Exam	In-person; Crowdmark Long answer, proof-based questions	To be centrally scheduled during the final exam period	Missed final exams are handled by Service Point
Homework	5 graded assignments, to be submitted on Crowdmark A new assignment will be posted roughly every two weeks All assignments will feature long-answer, proof-based questions. Your lowest of the 5 grades will be dropped.	Due dates are TBD, as class progresses.	Unexcused late assignments will not be accepted and will receive a grade of 0

Rules and Regulations:

- The purpose of the Policy on Assessment of Student Learning (PASL) is to provide a set of common principles to guide the assessment of students' learning. Per PASL, assessment should be equitable and consistent, and promote effective learning experiences, a healthy learning environment, and academic integrity. Learn more on the For Students page of the PASL website. Also see Faculty of Science-specific rules on the implementation of PASL.
- Legally mandated academic accommodations are handled by Student Accessibility and Achievement. For more information see <https://www.mcgill.ca/access-achieve/>
- Academic considerations for assessments that are missed or late for valid reasons will be provided at the instructor's discretion. As per the new Quebec guidelines, medical notes are not required for absences of less than 5 days. *Note that repeated, similar requests for academic considerations in this course are unlikely to be granted.*
- In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French written work that is to be graded. This does not apply to courses in which acquiring proficiency in a language is one of the objectives. (Approved by Senate on 21 January 2009)
Conformément à la Charte des droits de l'étudiant de l'Université McGill, chaque étudiant a le droit de soumettre en français ou en anglais tout travail écrit devant être noté, sauf

dans le cas des cours dont l'un des objets est la maîtrise d'une langue. (Énoncé approuvé par le Sénat le 21 janvier 2009)

- McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the Code of Student Conduct and Disciplinary Procedures (Approved by Senate on 29 January 2003) (See McGill's guide to academic honesty for more information).
- In the event of extraordinary circumstances beyond the University's control, the content and/or assessment tasks in this course are subject to change and students will be advised of the change.

Shortlist of Resources: The Department of Mathematics and Statistics, as well as the University at large, have many resources to help you succeed in this course and throughout your degree. A more extensive list of resources can be found in the MyCourses Webpage for this Course (under "Content"), as well as the Department's EOSW webpage. A shortlist of the most commonly used resources is available here:

- The Wellness Hub is a centralized website for student physical and mental health resources.
- The Math Help Desk is staffed by knowledgeable math students who can help answer your questions related to your courses. They have tutors on M-F from noon-5 PM in Burnside Hall room 911.
- Statistics Online for Students is a resource which offers online help on statistics courses in the department. Click on the link above and you will be added to a Microsoft Teams group where the schedule will be announced.
- SUMS is the Society of Undergraduate Mathematics Students. Join their Facebook group to get on their listserv, connect with other students in the department, and participate in some of their activities/social events.
- GSAMS is the Graduate Student Association for Mathematics and Statistics (GSAMS). All graduate students are represented by GSAMS, and they hold regular events throughout the semester to promote graduate student community.
- Advising is an important resource to guide you throughout the duration of your degree, and advisors can help with answering questions related to your degree program. Check out the Department's Advising Website to find out how to get in touch with a Departmental Advisor, should you need to do so.
- The Office for Mediation and Reporting is a McGill centralized office used to file a formal report of discrimination, harassment, or sexual violence; learn about policies and processes; or be connected to additional supports.
- The Office for Sexual Violence, Response, Support and Education provides support for all members of the McGill community who have been impacted by sexual violence (whether it be sexual harassment or assault, gender-based or intimate partner violence, or cyberviolence) and works to foster a culture of consent on campus and beyond.
- The Office of the Dean of Students (Case Management) is a collaborative process between a student, a Case Manager, and other concerned parties with the intention of improving the student's academic and personal outcomes. Case managers are trained in confidentiality, disclosures of sexual violence, and mental health first aid.